



StockAnalytix MVP

Standard Operating Procedure (SOP)

Operations · Research Workflow · Maintenance

Step-by-step procedures for setting up, running, using, maintaining and troubleshooting the StockAnalytix research platform.

1. Purpose & Scope

This SOP defines the standard procedures for operating StockAnalytix MVP — an AI-powered stock research platform. It covers setup, the daily research workflow, importing third-party reports, notes/tracking, AI configuration, maintenance/QA, troubleshooting and compliance guardrails.

Applies to

- Anyone running or maintaining the application locally.
- Analysts using the app for stock research and note-taking.
- Developers extending features (build/verify procedure).

Golden rule

StockAnalytix is research support only. It never gives buy/sell advice or guaranteed predictions. Every output carries a disclaimer and must be independently verified before any decision.

2. Setup & Startup Procedure

2.1 First-time setup

- 1 Open a terminal in the project folder.
- 2 Run ``npm install`` to install dependencies.
- 3 Open ``.env.local`` and paste the three AI keys (no space after ``=``): `GEMINI_API_KEY`, `GROQ_API_KEY`, `OPENAI_API_KEY`.
- 4 Save ``.env.local``.

2.2 Start the app

- 1 Development: run ``npm run dev`` ! open `http://localhost:3000`.
- 2 Production (faster, no recompile lag): run ``npm run build`` then ``npm start``.
- 3 After ANY change to ``.env.local``, restart the server so new keys load.

Key format check

Keys must be pasted directly after ``=`` with NO leading space. Gemini key worked despite an unusual prefix; Groq starts with ``gsk_``; OpenAI starts with ``sk-``. A space after ``=`` is the most common cause of 'AI disabled'.

3. Standard Research Workflow

The core daily procedure for researching a stock.

- 1 Go to Analyze Stock (sidebar) and search a symbol (e.g. AAPL, RELIANCE.NS). Pick market US / NSE / BSE in the top bar if needed.
- 2 Review the Overview, Technical, Fundamentals, Valuation and Risk tabs for the computed snapshot.
- 3 Open the Momentum tab for the momentum score, graphs, scenarios and (optionally) start a 15-Day Watch.
- 4 Open the Evaluation tab for the CAN SLIM checklist, composite rating, Acc/Dis & SMR grades and Alpha/Beta.
- 5 Open the Analytics tab for seasonality and the historical backtest signal-edge.
- 6 Open the Research tab to generate the full CANSLIM research note; export to Excel/PDF as needed.
- 7 Use the Notes tab to record text/voice observations for this stock.
- 8 Run AI deep dives where useful (Momentum Deep Dive, AI Stock Evaluation, AI Research Note).

Research language only

When interpreting outputs, use research framing — 'momentum improving', 'watch for confirmation', 'extended', 'avoid chasing'. Never read any output as a recommendation to buy or sell.

4. Import Report Procedure

Use this to make a static third-party report (MarketSmith / Market Mojo / broker) LIVE and actionable. Only import reports you are entitled to (your own subscription). Do NOT scrape paywalled sites.

- 1 Open Import Report (sidebar).
- 2 Paste the report (HTML or text) OR upload a .html / .txt file.
- 3 Click 'Analyze Report' (allow 30–60s — it runs AI extraction + live cross-check).
- 4 Read the Report Health Check: age, price move since the report, and the AI 'Is this still valid?' verdict.
- 5 Review the Cross-Check (report vs our live data) and 'What Our Analysis Adds' (backtest, seasonality, alpha/beta).
- 6 Use 'View Original Report' to see the rendered source alongside the analysis.
- 7 Optionally: 'Save Key Points to Notes', 'Track this Thesis' (alerts + 15-day watch), or chat with the report.
- 8 The imported data is auto-saved and appears as a banner on that stock's Research tab.

5. Notes & Tracking Procedure

5.1 Notes (text & voice)

- 1 On any stock's Analyze page, open the Notes tab (or use Master Notes in the sidebar).
- 2 Type a note (Cmd/Ctrl+Enter to save) or click 'Voice Note' to record (max ~90s).
- 3 All notes are saved per stock with a timestamp and viewable in Master Notes.

5.2 15-Day Watch

- 1 On the Momentum tab, click 'Start 15-Day Momentum Watch' to save a baseline.
- 2 Re-open later to see baseline-vs-current changes and run 'Analyze What Changed'.

Voice note storage

Voice notes are stored in the browser (localStorage), capped ~90s for size. Microphone permission is required on first use. Clearing browser data removes notes.

6. AI Provider Configuration

All AI features route through a single resilient client with tiered routing and automatic fallback.

Task type	Provider order
Reasoning-heavy (research note, evaluation, import cross-check, Research note)	OpenAI ! Gemini ! Groq
Light (chat, news summaries)	Groq ! Gemini ! OpenAI

- Each provider retries transient errors (503 / overload / 429) and falls back across models, then to the next provider.
- Optional model overrides: GEMINI_MODEL, GROQ_MODEL, OPENAI_MODEL (e.g. set OPENAI_MODEL=gpt-4o for stronger reasoning).
- Web-grounded deep research uses Gemini Google Search (quota-limited on free keys; degrades gracefully).

7. Maintenance & QA Procedure

7.1 Before considering any change 'done'

- 1 Run ``npx tsc --noEmit`` — confirm no NEW type errors over the baseline (the project tolerates a known baseline via `ignoreBuildErrors`).
- 2 Run ``npx next build`` — confirm all pages generate successfully.
- 3 If a dev server is running, stop it before building (dev + build share the `.next` folder).

7.2 Functional QA

- 1 Open QA / Diagnostics (sidebar) and click 'Run Tests'.
- 2 All Route, Momentum, Evaluation, Analytics and Timeframe tests should Pass.
- 3 Provider-missing data appears as Warning (not Fail) — this is expected and acceptable.

Warnings are OK

Some QA items are Warnings by design (e.g. a newly-listed stock with <1 month of hourly data, or a legacy symbol). These reflect honest data limitations, not bugs.

8. Troubleshooting Guide

Symptom	Action
AI 'disabled' / not working	Check .env.local keys have NO space after '='; restart server.
AI 503 'high demand'	all busy. led automatically (retry + Gemini! Groq / OpenAI)
Fundamentals show 'Data unavailable'	Data is not available. Yahoo; some India-specific fields (FII/DII) are
PDF was blank	Resolved. PDF is built programmatically (jsPDF), not a DOM
Analyze is slow / 50s+	First call after a code change recompiles (dev only). Warm calls are 2–4s.
Import report timed out	It runs 2 AI calls + full compute; allow 30–60s. Retry once.
Build fails on /api/.. page data	A dev server is conflicting; stop it, delete .next, rebuild.

9. Compliance & Guardrails

Always

- Use research language only — no buy/sell/hold recommendation.
- Keep the mandatory disclaimer on every report and AI output.
- Show a clean 'Data unavailable' state with a reason when a value is genuinely missing — never invent data.
- Only import reports you are entitled to (your own subscription). No scraping of paywalled/login sites.

Never

- Never present any output as a guaranteed prediction or price target recommendation.
- Never fabricate proprietary third-party ratings — show our own computed equivalents, clearly labelled.
- Never store third-party API keys in client code or commit .env.local.

10. Quick Reference

Action	Where
Analyze a stock	Sidebar ! Analyze Stock ! search symbol
Momentum score & watch	Analyze ! Momentum tab
CAN SLIM evaluation	Analyze ! Evaluation tab
Backtest & seasonality	Analyze ! Analytics tab
Full CANSLIM research note	Analyze ! Research tab
Notes (text/voice)	Analyze ! Notes tab, or Master Notes
Import a 3rd-party report	Sidebar ! Import Report
Ask AI about a stock	Sidebar ! AI Research Chat
Run health checks	Sidebar ! QA / Diagnostics

StockAnalytix MVP — SOP v1.0

Research support only. Not buy/sell advice. No guaranteed prediction. Always verify independently.